The Autoencoder Paradigm Shift

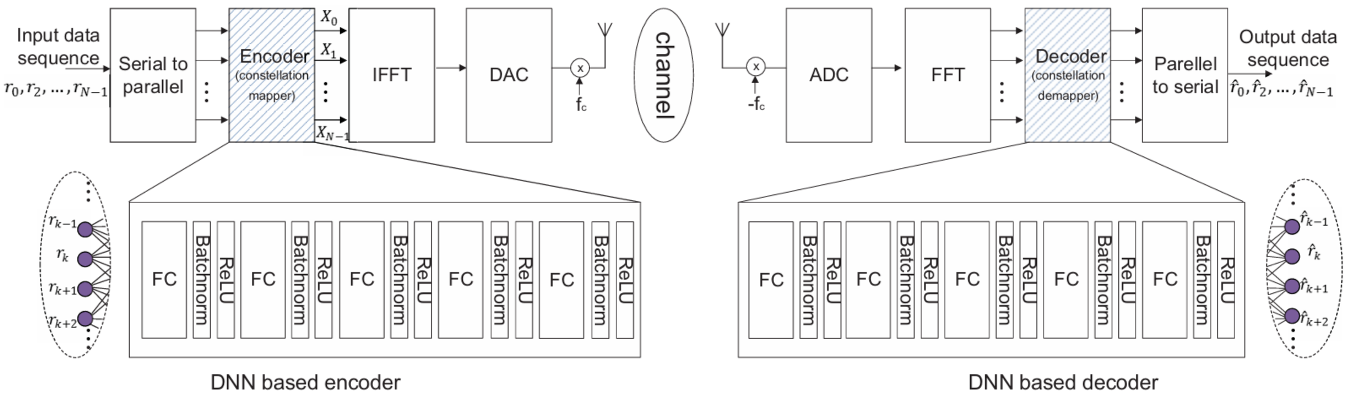
Introduced by Kim et al., **PRNet** is the first deep learning-based system for joint PAPR and BER optimisation. It treats the entire transceiver chain as a single deep **autoencoder**:

- **Encoder** $\longrightarrow$ Neural Network Transmitter
- **Decoder** $\longrightarrow$ Neural Network Receiver
- **Bottleneck** $\longrightarrow$ Noisy wireless channel

Both the transmitter and receiver are jointly trained, allowing the network to:

- Learn a custom, data-dependent constellation mapping that jointly minimises **BER** and **PAPR**.
- Eliminate side information (SI) overhead entirely — the decoder implicitly learns the encoder's mapping.
- Bypass the exponential search spaces of classical methods (e.g. SLM).

PRNet Transceiver Architecture



Standard modulation and demodulation are completely replaced by a **DNN encoder and decoder** sharing a symmetric topology: a **128-node input layer** (I/Q representation) feeding into 5 hidden layers (2048 nodes each) and a **128-node output layer**. Each hidden sub-block applies **FC** $\rightarrow$ **Batch Norm** $\rightarrow$ **ReLU**:

- **FC:** Dense connectivity enables joint subcarrier processing.
- **BN:** Stabilises training to prevent gradient collapse.
- **ReLU:** Provides non-linearity for non-convex PAPR–BER mappings.

*Note: The final output layer is **strictly linear** (no BN/ReLU) to output unbounded I/Q spatial coordinates (detailed on next slide).*

End-to-End System Model

1. **Input Representation:** Real I/Q vector

$$\boldsymbol{r} = [I_0,\, Q_0,\, \ldots,\, I_{N-1},\, Q_{N-1}]^T \in \mathbb{R}^{2N}$$

2. **DNN Encoder:** f parameterized by weights θ_f

$$\boldsymbol{X} = f(\boldsymbol{r};\, \theta_f) \in \mathbb{C}^N$$

3. **OFDM Mod & Channel:** IFFT, then Rayleigh fading $\mathbb{H}$ & AWGN

$$y[n] = \mathbb{H}\big(\mathrm{IFFT}(\boldsymbol{X})\big)$$

4. **Demod & EQ:** FFT to frequency domain, zero-forcing

$$\hat{Y}_k = X_k + \frac{E_k}{H_k}$$

5. **DNN Decoder:** g parameterized by weights θ_g

$$\hat{\boldsymbol{r}} = g(\hat{\boldsymbol{Y}};\, \theta_g) \in \mathbb{R}^{2N}$$

6. **Fully Differentiable End-to-End Chain:**

$$\hat{\boldsymbol{r}} = g \circ \mathrm{FFT} \circ \mathbb{H} \circ \mathrm{IFFT} \circ f(\boldsymbol{r})$$

Why I/Q?

Unlike categorical one-hot encoding, continuous I/Q decomposition strictly halves the input dimension (128 vs. 256), allows the network to optimise continuous spatial coordinate geometry directly, and enables explicit per-symbol power normalisation ($\mathbb{E}[|X|^2] = 1$).

Joint Optimization & Multi-Objective Loss

Reconstruction loss (MSE in $\mathbb{R}^{2N}$):

$$\mathcal{L}_1(\boldsymbol{r}, \hat{\boldsymbol{r}}) = \frac{1}{2N}\Big\|\boldsymbol{r} - g\Big(\mathrm{FFT} \circ \mathbb{H} \circ \mathrm{IFFT}\Big(f(\boldsymbol{r};\, \theta_f)\Big);\, \theta_g\Big)\Big\|_2^2 = \frac{1}{2N}\sum_{i=0}^{2N-1}\Big(r_i - \hat{r}_i\Big)^2$$

PAPR penalty (oversampled time-domain waveform, $L \geq 4$):

$$\mathcal{L}_2(\boldsymbol{r}) = \mathrm{PAPR}\Big\{\mathrm{IFFT}\Big(f(\boldsymbol{r};\, \theta_f)\Big)\Big\} = \frac{\max\limits_{0 \leq n \leq NL-1} |x[n]|^2}{\frac{1}{NL}\sum\limits_{n=0}^{NL-1} |x[n]|^2}$$

Joint loss: (λ governs the BER–PAPR trade-off)

$$\mathcal{L}(\boldsymbol{r}, \hat{\boldsymbol{r}}) = \mathcal{L}_1(\boldsymbol{r}, \hat{\boldsymbol{r}}) + \lambda\, \mathcal{L}_2(\boldsymbol{r})$$

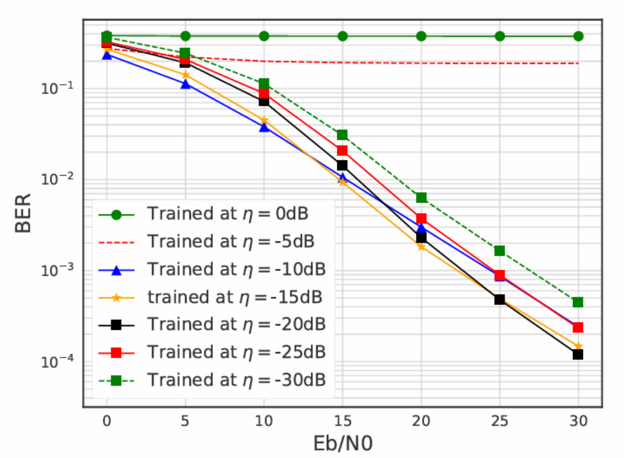
Why Regression MSE?

The original PRNet formulation uses **Cross-Entropy loss** for categorical, one-hot encoded inputs. Because our system adopts a **continuous I/Q parameterization**, the reconstruction task becomes a regression problem. Therefore, we utilize **Mean Squared Error (MSE)** to compute the loss in the $2N$-dimensional real-valued signal space.

Two-Stage Training Methodology

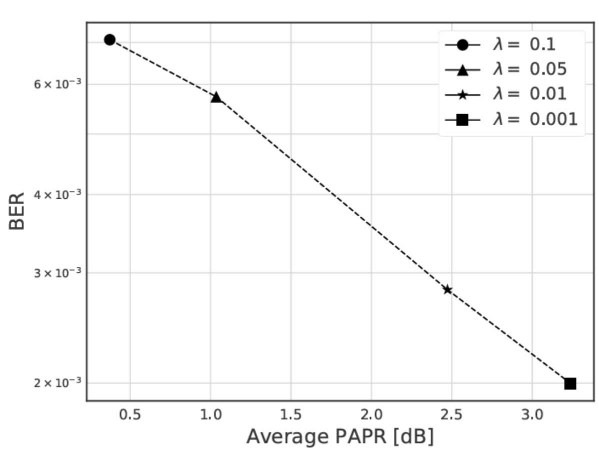
Denoising autoencoder: Artificial noise at a corruption level η (noise-to-signal ratio) is injected during the forward pass to simulate the physical channel and force the decoder to learn robust decision boundaries.

Stage 1 — Noise Selection ($\lambda = 0$): Sweep η while training with reconstruction loss $\mathcal{L}_1$ only.



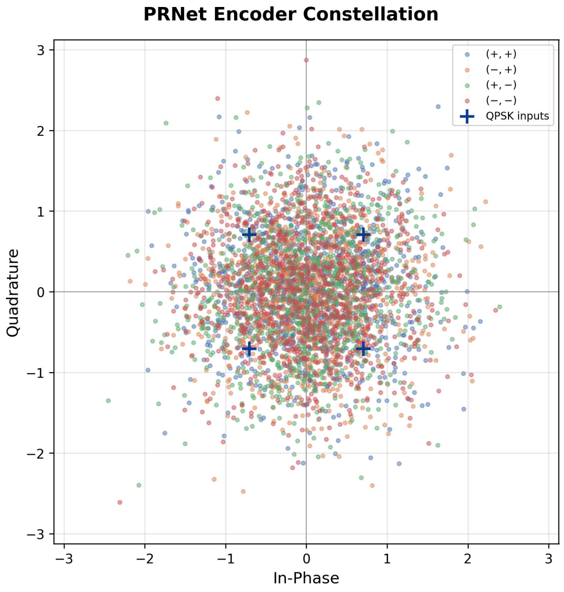
For excessive corruption levels ($\eta = 0$ and -5 dB), the model fails to learn meaningful boundaries entirely, while at very low corruption levels ($\eta = -30$ dB) it overfits to the noiseless scenario. The best point is $\eta^\star = -15$ dB; any divergence from this optimal point steadily increases the overall BER.

Stage 2 — Trade-Off Training ($\eta = \eta^\star$): Sweep penalty weight λ under the full loss $\mathcal{L} = \mathcal{L}_1 + \lambda\,\mathcal{L}_2$.



As λ increases, PAPR decreases monotonically while the BER increases. Heavy penalties ($\lambda = 0.1$) aggressively squash the waveform but severely compromise the error rate, whereas minimal penalties ($\lambda = 0.001$) prioritize low BER at the expense of higher PAPR. The ideal operating point is $\lambda = 0.01$, achieving meaningful PAPR reduction without severely degrading the BER baseline.

Learned Constellation Shaping

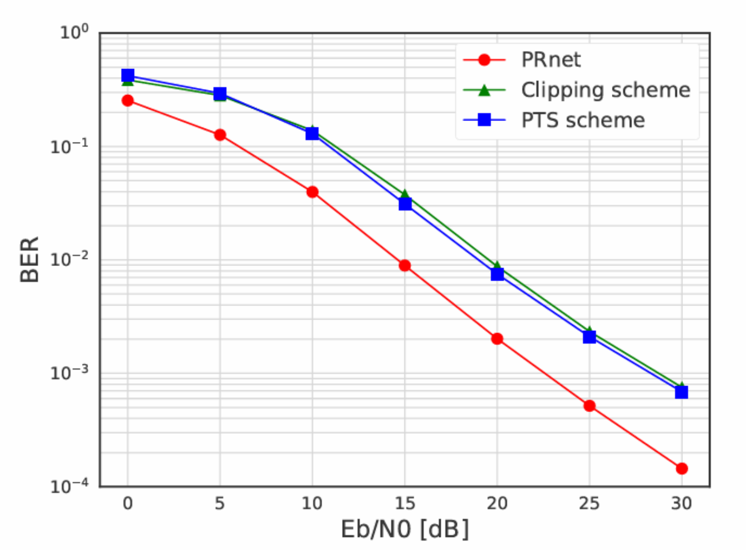


Unlike classical OFDM with rigid QPSK boundaries, the PRNet encoder **dynamically disperses** constellation points:

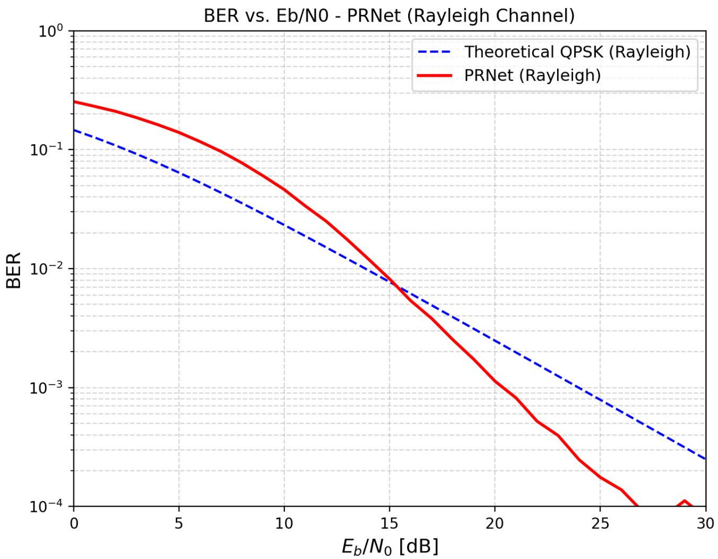
- Maps discrete input data into a **continuous, cloud-like** spatial distribution.
- Trades traditional Euclidean spacing for optimal PAPR reduction.
- The jointly trained decoder learns to recover data from this custom distribution without explicit knowledge of the mapping.

BER Performance: Validation

Original Paper



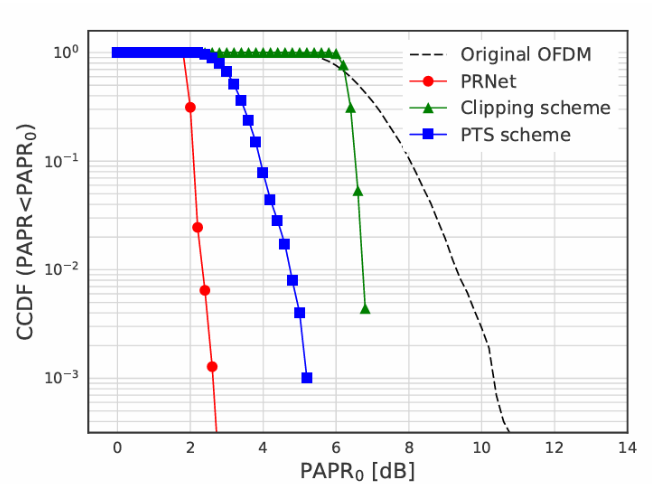
Our Reproduction



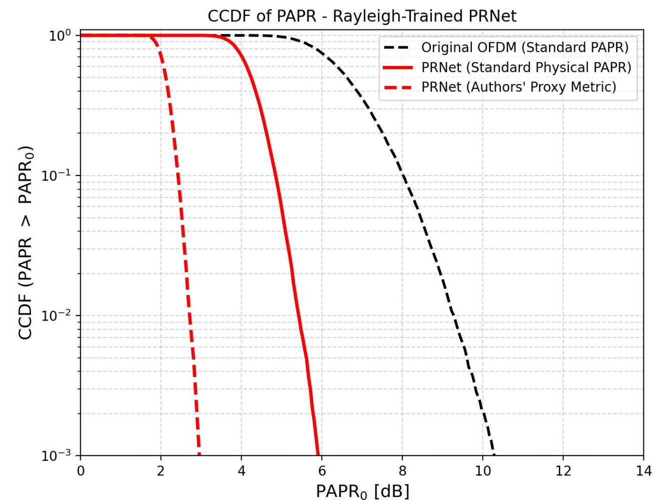
- PRNet **outperforms** the theoretical QPSK baseline — learns an implicit coding gain under the Rayleigh fading channel.
- Deterministic inference: single forward pass, no side information required.

PAPR Reduction & Proxy Metric Discrepancy

Original Paper



Our Reproduction



Metric Mismatch: The authors used an amplitude-based proxy instead of physical power:

- **Authors' proxy:** $\mathrm{PAPR}_{\mathrm{proxy}}\,[\mathrm{dB}] = 10\log_{10}(\max_n |x_{\mathrm{norm}}[n]|) \Rightarrow$ Reports 3 dB
- **Standard metric:** $\mathrm{PAPR}_{\mathrm{std}}\,[\mathrm{dB}] = 10\log_{10}(\max_n |x_{\mathrm{norm}}[n]|^2) = 2 \times \mathrm{PAPR}_{\mathrm{proxy}}\,[\mathrm{dB}] \Rightarrow$ Actually 6 dB

Despite this $2\times$ discrepancy, PRNet still achieves a **substantial** 4**–**5 **dB reduction** vs. original OFDM (≈ 10.5 dB).

Computational Complexity & Limitations

Complexity Correction:

- Original claim: $\mathcal{O}(L \cdot K \cdot M)$.
- **True complexity:** $\mathcal{O}(L \cdot K^2)$ — hidden-layer matrix multiplications ($K = 2048$) dominate; independent of M.
- GPU inference: $480\ \mu$s vs. PTS: $2{,}890\ \mu$s ($\approx 6\times$ speedup).

Practical Limitations of PRNet:

- **Scalability:** FC layers scale as $\mathcal{O}(N^2)$ — impractical for commercial $N = 1024$.
- **Channel mismatch:** Overfits to the training fading model; degrades under unmodelled conditions.
- **Configuration rigidity:** Changing N or M requires full architectural redesign and retraining.
- **Interpretability:** Black-box weights prevent worst-case guarantees required by telecom standards.